

Study on the preparation of wood vinegar from biomass residues by carbonization process

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panelQiaomei Wu, [Shouyu Zhang](#), Baoxin Hou, Hongjun Zheng, Wenxiang Deng, Dahai Liu, Wenjiao Tang

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Highlights

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Cotton stalk was used to produce wood [vinegar](#) by [carbonization](#) process.

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The highest yield of the refined wood [vinegar](#) reaches 25% at 350 °C.

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The main organic components in the wood vinegars are acids, [phenols](#) and ketones.

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The presence of [potassium](#) restrains the formation of ketones.

Abstract

In the paper, the production of wood vinegar from Chinese fir sawdust (FS), cotton stalk (CS) and bamboo sawdust (BS) by [carbonization](#) process was addressed. The wood vinegar yield was investigated and the organic compounds contained were determined by gas chromatography and mass spectrometry. It was found that the refined wood vinegar yield of FS increased firstly and then decreased with increasing carbonization temperature and the highest yield reached about 25% in 350–450 °C. The relative contents of acids and ketones from FS decreased and that of [phenols](#) increased with increasing temperature. The relative contents of acids and phenols in the wood vinegars produced from the samples were in the order of BS > CS > FS and that of ketones reversed. KCl solution treatment caused a decrease in the relative contents of the phenols and ketones, but an increase in that of the acids in FS wood vinegar.

Introduction

Wood vinegar (WV) or Pyroligneous acid (PA) is an acidic reddish-brown aqueous liquid. It is obtained by clarifying the liquid product of the carbonization process of woods or wood residues from wood processing industry, tree branches, bamboo, crop straw, fruit shell and other biomaterials (Wei et al., 2010). WV is often used as insect repellent, odor remover, wood preservative, soil or foliar fertilizer and plant growth promoter or inhibitor, animal feed additive, coagulant in NR sheets' preparation. Until now, lots of the products have been applied in a wide variety of markets (Wu et al., 2014, Shiny and Remadevi, 2014, Mu et al., 2004, Ferreira et al., 2005, Baimark and Niamsa, 2009).

The properties and utilizations of wood vinegar have been widely researched and the utilization is closely related to its compositions. It is reported that the high concentration of the organic acids in the WV exhibited the high antimicrobial activity (Ma et al., 2011) and the phenolic compounds showed significant antioxidant activity, and demonstrated dosage dependency (Loo et al., 2007, Loo et al., 2008). Wood vinegar contains lots of very complicated organic components and over 200 compounds have been found in the wood vinegar obtained from different resources (Guillén et al., 2001). In China, there have been some researches only on the organic components contained in the wood vinegars obtained directly from the charcoal kiln or WV Company. The wood vinegars are produced from Mongolian oak, birch, apple tree and pear by carbonization process.

Based on the heating rate, pyrolysis process can be divided into three modes: fast, intermediate and slow. As a kind of pyrolysis technology with low heating rate, carbonization process has been widely used in the conversion process of agricultural and forestry residues into high added-value products (Xiong et al., 2014). During the carbonization process, the raw material is heated up in oxygen-free environment and volatile matters emits from the parent materials and the three-phase matters of charcoal, liquid products, and fuel gas are produced. As a high application value product, WV can be obtained by clarifying from the liquid products. During the process, the final carbonization temperature employed plays a notable role in the distribution of the three-phase products (Kwapinski et al., 2010, Angin, 2013).

The production and characterization of wood vinegar from rubberwood was studied by Ratanapisit et al. (2009) and it was found that the optimum condition of pyrolysis was obtained at the heating rate of 1.4 °C/min to the final temperature of 550 °C with a WV yield of 27.45%. The main ingredient contained in the WV was acetic acid and included the high content of methanol compared to the WV produced from other woods. Moreover, Şensöz (2003) also found that there was an obvious difference in the ingredients of the wood vinegar produced from different raw materials and under different process conditions. However, there have been very few papers reported about WV preparation and the

influence of preparing condition and feedstock on the yields and compositions of the resultant wood vinegar.

In the paper, as agricultural and forestry residues, Chinese fir sawdust, cotton stalk and bamboo sawdust were chosen as the raw materials for the wood vinegar production. Because that the wood residue including the fir sawdust is the main biomaterials for the production of the wood vinegar (Wei et al., 2010), Chinese fir sawdust was chosen as main research object. The experiment was conducted on a vertical tubular carbonization reactor and the organic constituents of the resultant wood vinegar were analyzed and determined. Thus, the effect of carbonization temperature and feedstock type on the yields and organic constituents of the resultant wood vinegar could be investigated in detail. Furthermore, Chinese fir sawdust was pretreated by KCl solution and the resultant sample was carbonized in order to investigate the effect of the presence of potassium on the compositions of the resultant wood vinegar. The aim of the research was to provide a theoretical basis for the further development and utilization of WV.

Materials

Three agricultural and forestry residue samples, Chinese fir sawdust (FS), cotton stalk (CS) and bamboo sawdust (BS) were chosen as the raw materials for the investigation. Their proximate and ultimate analyses are shown in Table 1 and their ash analyses are listed in Table 2. The ash constituents of FS could not be determined due to its very low ash yield as shown in Table 1. Thus, the influence of the mineral matters contained in the biomass is slight and could be neglected. The samples were

Effect of carbonization temperature on FSWV yield

FS consists of celluloses, hemicelluloses and lignins with the content of 47.40%, 10.65% and 32.94%, respectively (Zhang et al., 2004). Fig. 2 shows the yields of the crude and refined WV produced from FS under the different carbonization temperature. As shown from the figure, the crude FSWV yield increases with the increasing carbonization temperature. The increase in the crude FSWV yield with the increasing carbonization temperature is mainly attributable to the degradation of hemicelluloses,

Conclusion

The yield of the crude WV increases with the increasing carbonization temperature and that of the refined WV increases firstly then decreases. The refined wood vinegar yield of 25% can be obtained in the carbonization temperature range of 350–450 °C. The relative contents of the acids and ketones contained in FSWV decrease and that of the phenols increases with the increasing temperature. The typical components with high content are the acids and phenols in BSWV, the ketones in FSWV and alcohols

Acknowledgement

This work was financially supported by the National Program on Key Basic Research Project (973 Program) of China (No. 2012CB214900).